

Introduction to Programming

Team Assignment | Telegram Bot

May 11, 2026

Telegram Bot for Programming Practice and File-Based Data Processing

The bot combines several programming concepts studied during the course: Telegram bot handlers, file processing, ZIP archives, dictionaries, pandas data analysis, simple calculations, timers, stickers, and location handling.

The bot can:

- process ZIP files
- search information inside files
- work with datasets
- create simple charts
- use calculator and timer functions
- send stickers and handle location

Functionality:

The bot uses a button-based menu to make interaction easier for users. The user can select options instead of typing long commands manually.

Telegram bot name:

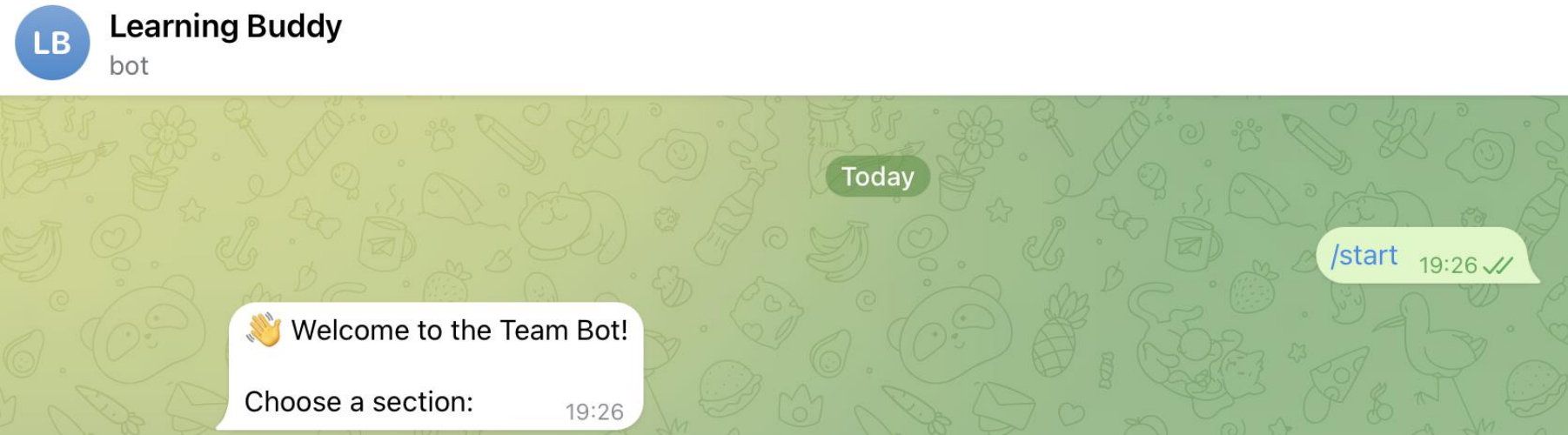
Learning Buddy

Telegram bot username:

@MNU_learningbuddy_bot

Main Menu and Navigation

The menu consists of three dimensions – each one designed for different purposes and consists of different functions.



*To begin working with bot – write down **'/start'***



Fun Work



Data Analytics



ZIP Search

Three dimensions designed by each team participant:

- **Fun Work**
- **Data Analytics with CSV File**
- **ZIP Search**

Zip File Working Part

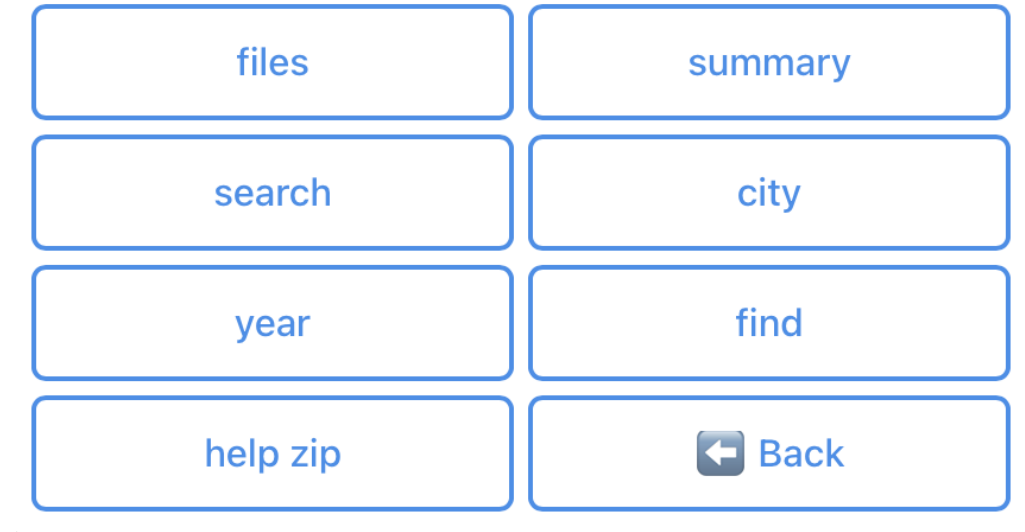
The user can upload a .zip archive. The bot saves the file and remembers it for the current user.

Programming concepts used:

- document handler
- file downloading
- zipfile
- dictionary user_zip_files
- try/except



The ZIP file can be worked through different functions



Files Inside ZIP

Functionality:

The bot opens the uploaded archive and displays all files stored inside it.

Usability:

The user can quickly check the structure of the uploaded ZIP file before searching inside it.

files 20:25 ✓✓

Files inside ZIP

1.  ip_data_2020.txt
2.  ip_data_2021.txt
3.  regions.csv

20:25

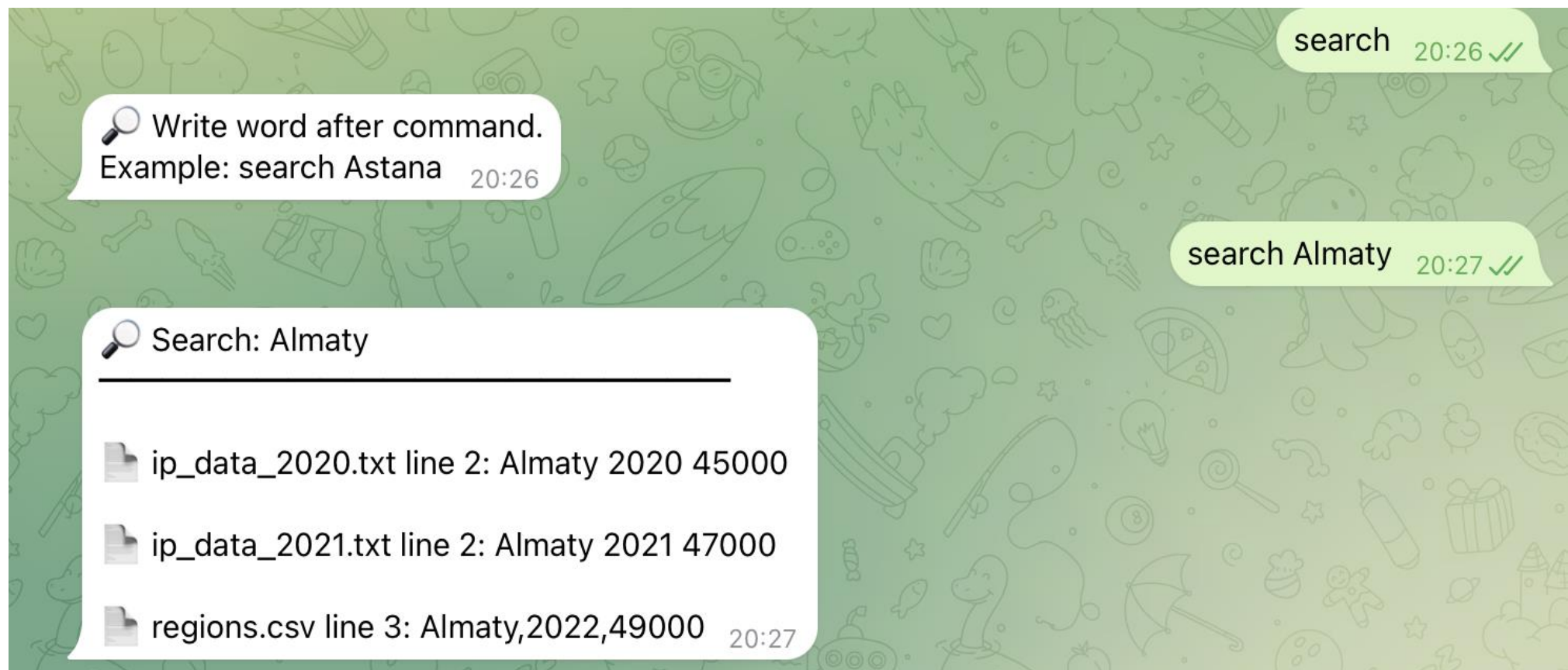
Search Word in ZIP

Functionality:

The bot searches for a word in all .txt and .csv files inside the ZIP archive.

Example:

If the user writes search Astana, the bot finds all lines where this word appears.



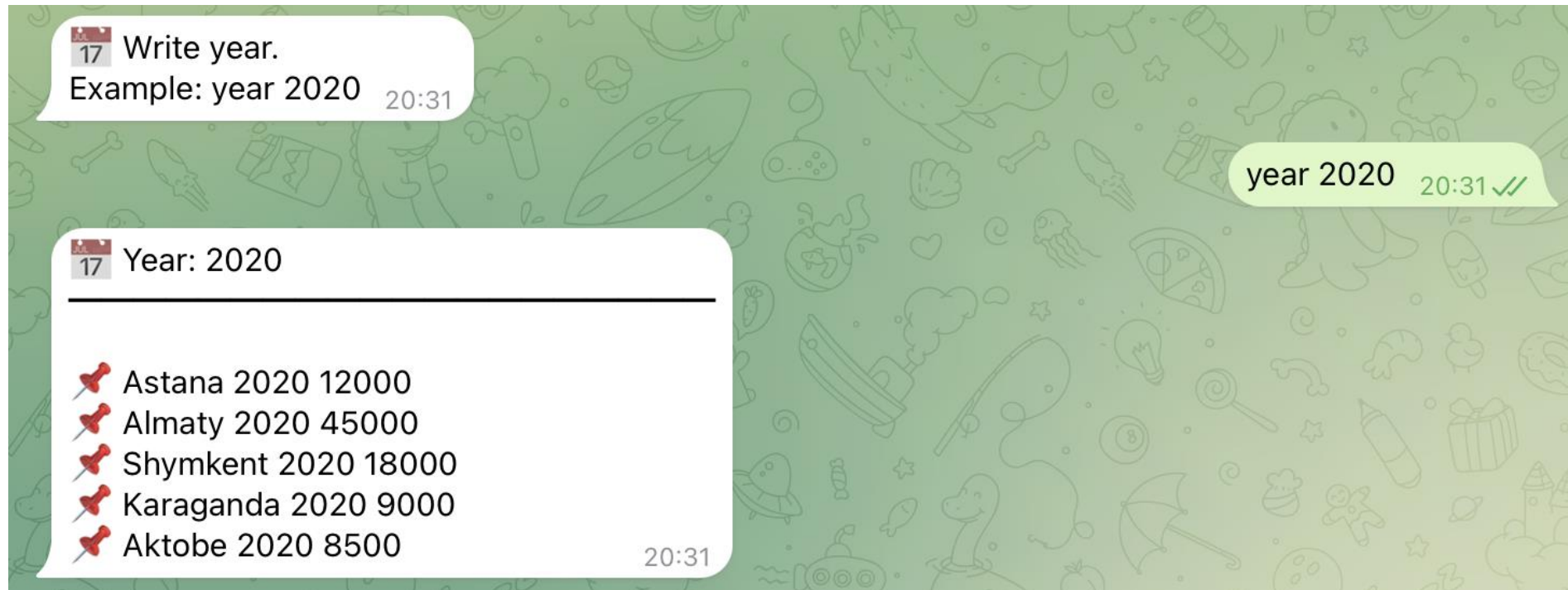
Search by Year

Functionality:

The bot searches all records connected with a selected year.

Example:

The command year 2020 returns all lines that contain data for 2020.



Find by City and Year

Functionality:

This function combines two conditions: city and year.

Example:

The user writes find Astana 2021, and the bot returns only records where both values are found.



ZIP Summary

Functionality:

The bot gives a short summary of the archive.

It shows:

- total number of files;
- number of .txt files;
- number of .csv files;
- total number of lines.

summary 20:38 ✓✓



Archive summary



All files: 3



TXT files: 2



CSV files: 1



Total lines: 19

20:38

ZIP help

If person needs any type of help in understanding of functions represented by ZIP dimension – ZIP help button shows commands description



ZIP SEARCH COMMANDS

files - list files in ZIP

summary - archive stats

search <word> - search word in all files

city <name> - search by city

year <year> - search by year

find <city> <year> - search by city + year

First send a ZIP file, then use commands. 20:40

help zip 20:40 ✓✓

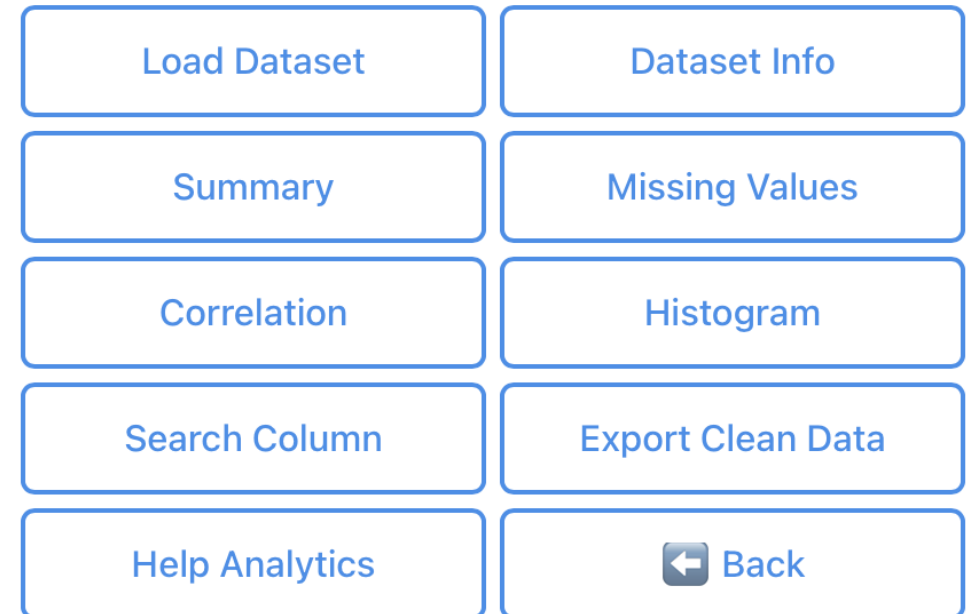
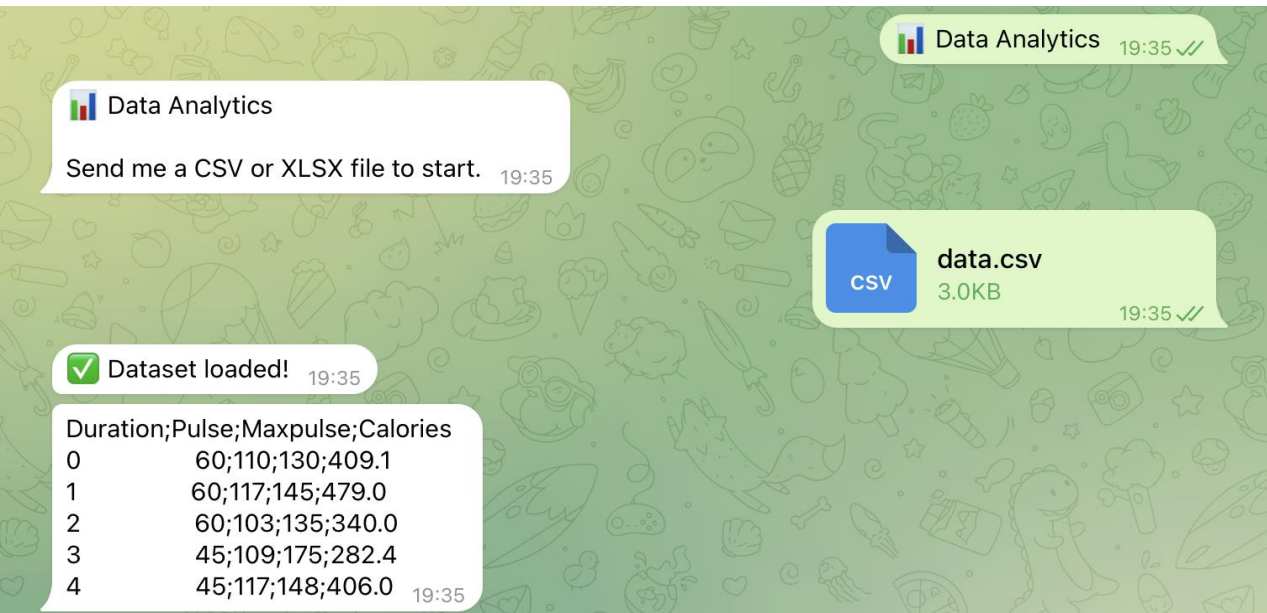
Data Analytics with CSV

The user can upload a .csv archive. The bot saves the file and remembers it for the current user.

Functionality:

The bot provides a separate menu for data analysis. Instead of typing long commands, the user can press buttons.

The CSV file can be worked through different functions



Load Dataset

Functionality:

This function allows the user to upload a dataset file. After uploading, the bot reads the file and saves the dataset for the current user.

Example:

The user presses Load Dataset, then sends a .csv file. The bot reads it and shows that the dataset was loaded successfully.

The screenshot shows a Telegram chat interface with a green background and various icons. The chat history includes:

- A user message: "Send a CSV or XLSX file." (19:54)
- A bot response: "Load Dataset" (19:54) with a green checkmark.
- A user message: "support2.csv" (3.0MB) (19:54) with a green checkmark.
- A bot response: "Dataset loaded!" (19:54) with a green checkmark.

The bot's response displays the following dataset:

	age	death	sex	hospdead	slos	d.time	dzgroup \
1	62.84998	0	male	0	5	2029	Lung Cancer
2	60.33899	1	female	1	4	4	Cirrhosis
3	52.74698	1	female	0	17	47	Cirrhosis
4	42.38498	1	female	0	3	133	Lung Cancer
5	79.88495	0	female	0	16	2029	ARF/MOSF w/Sepsis

	dzclass	num.co	edu ...	crea	sod	ph	glucose \
1	Cancer	0	11.0 ...	1.199951	141.0	7.459961	NaN
2	COPD/CHF/Cirrhosis	2	12.0 ...	5.500000	132.0	7.250000	NaN
3	COPD/CHF/Cirrhosis	2	12.0 ...	2.000000	134.0	7.459961	NaN
4	Cancer	2	11.0	0.799927	139.0	NaN	NaN

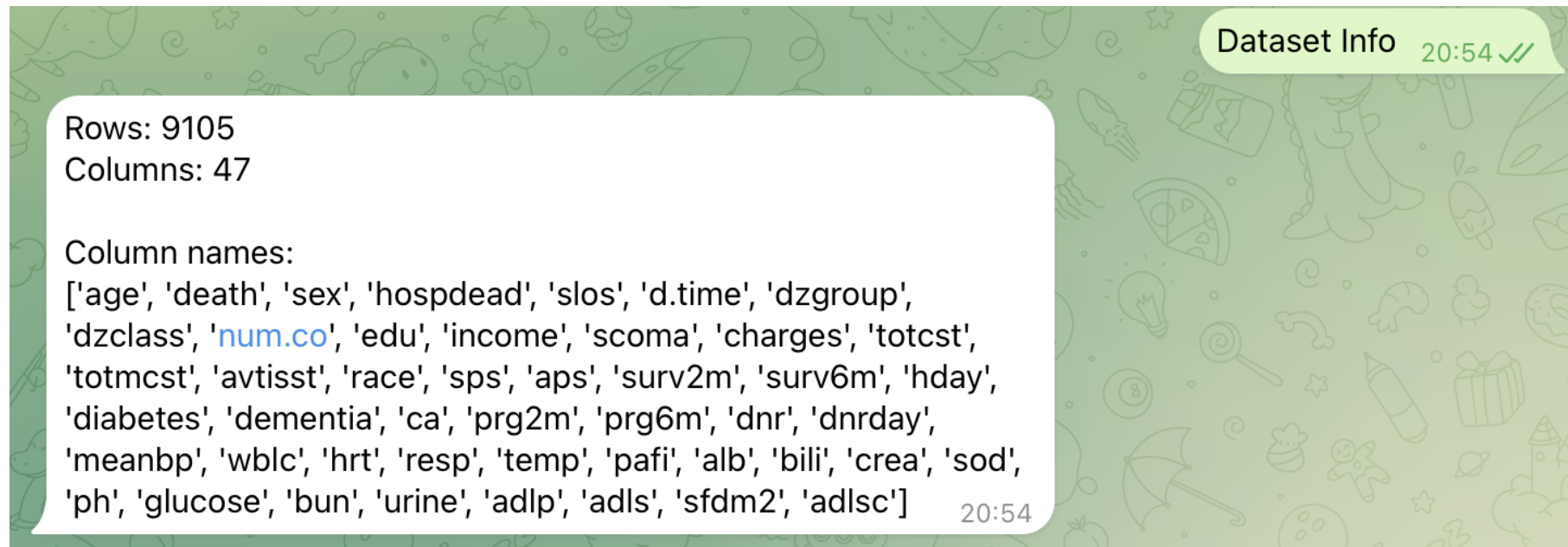
Dataset Info

Functionality:

This function shows basic information about the uploaded dataset.

The bot can show:

- number of rows;
- number of columns;
- column names;



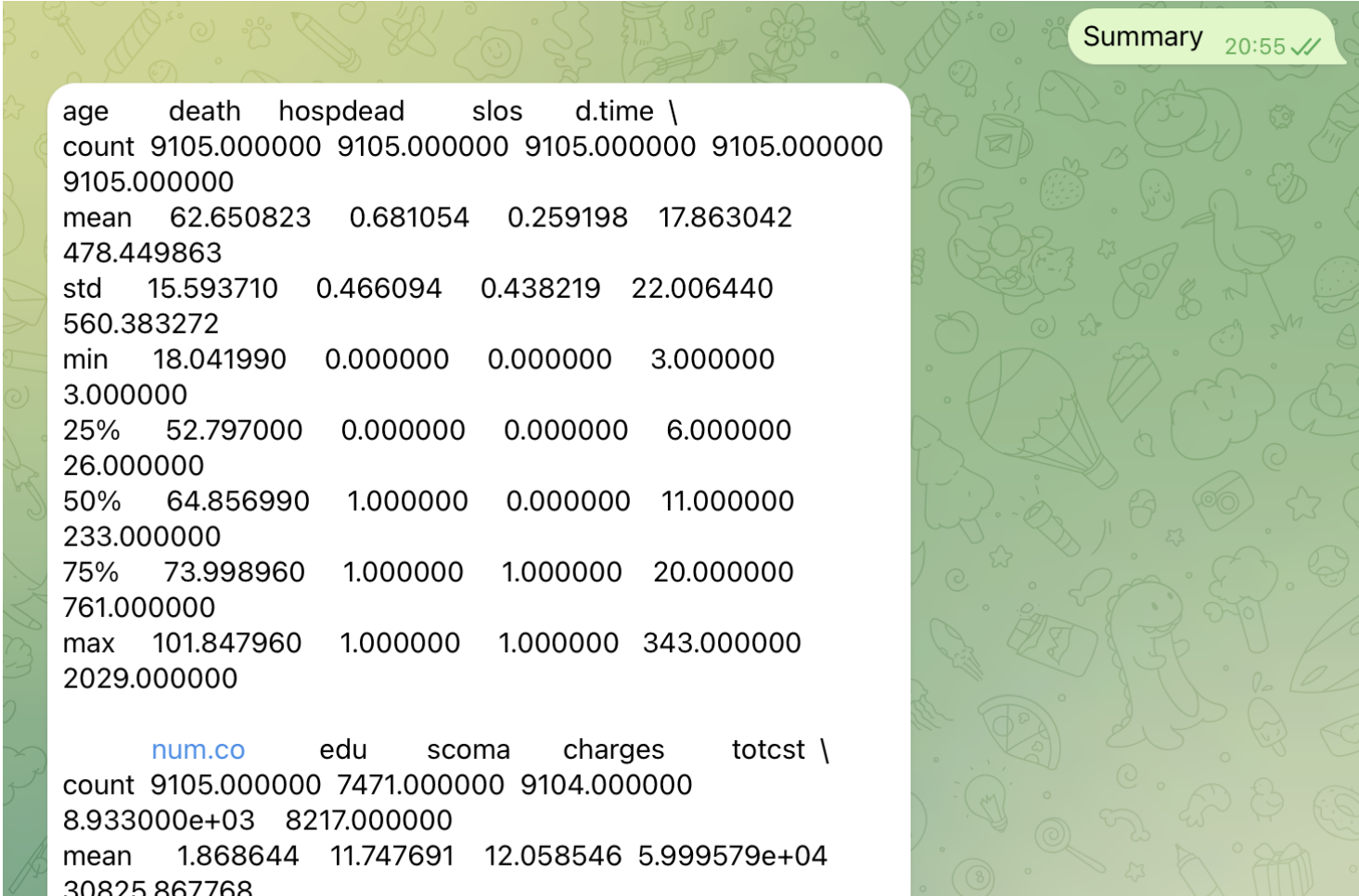
Dataset Summary

Functionality:

This function shows basic statistical information about numeric columns in the dataset.

The bot can show:

- count;
- mean;
- standard deviation;
- minimum value;
- maximum value;
- quartiles.



```
age    death  hospdead    slos    d.time \
count  9105.000000  9105.000000  9105.000000  9105.000000
9105.000000
mean   62.650823   0.681054   0.259198   17.863042
478.449863
std    15.593710   0.466094   0.438219   22.006440
560.383272
min    18.041990   0.000000   0.000000   3.000000
3.000000
25%    52.797000   0.000000   0.000000   6.000000
26.000000
50%    64.856990   1.000000   0.000000   11.000000
233.000000
75%    73.998960   1.000000   1.000000   20.000000
761.000000
max    101.847960   1.000000   1.000000   343.000000
2029.000000

      num.co    edu    scoma    charges    totcst \
count  9105.000000  7471.000000  9104.000000
8.933000e+03  8217.000000
mean    1.868644  11.747691  12.058546  5.999579e+04
30825 867768
```

Dataset Missing Values

Functionality:

This function checks whether the dataset contains empty or missing values.

The bot can show:

- which columns have missing values;
- how many missing values each column has;
- whether the dataset is complete or not.

age	0
death	0
sex	0
hospdead	0
slos	0
d.time	0
dzgroup	0
dzclass	0
num.co	0
edu	1634
income	2982
scoma	1
charges	172
totcst	888
totmcst	3475
avtisst	82
race	42

Missing Values 20:58 ✓✓

Dataset Correlation

Functionality:

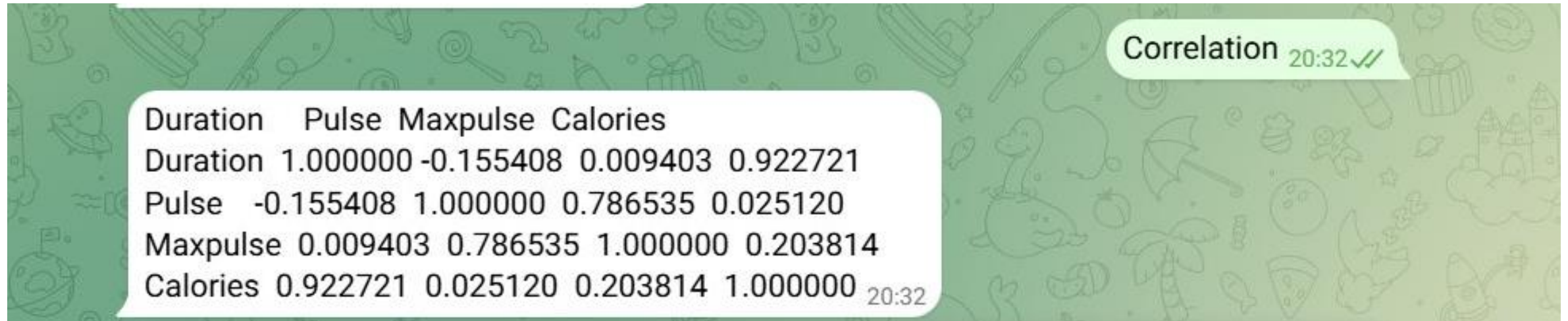
This function shows the relationship between numeric columns in the dataset.

What it does:

The bot calculates correlation values between numeric columns and sends the result to the user.

Why it is useful:

Correlation helps understand whether two numeric variables move together.



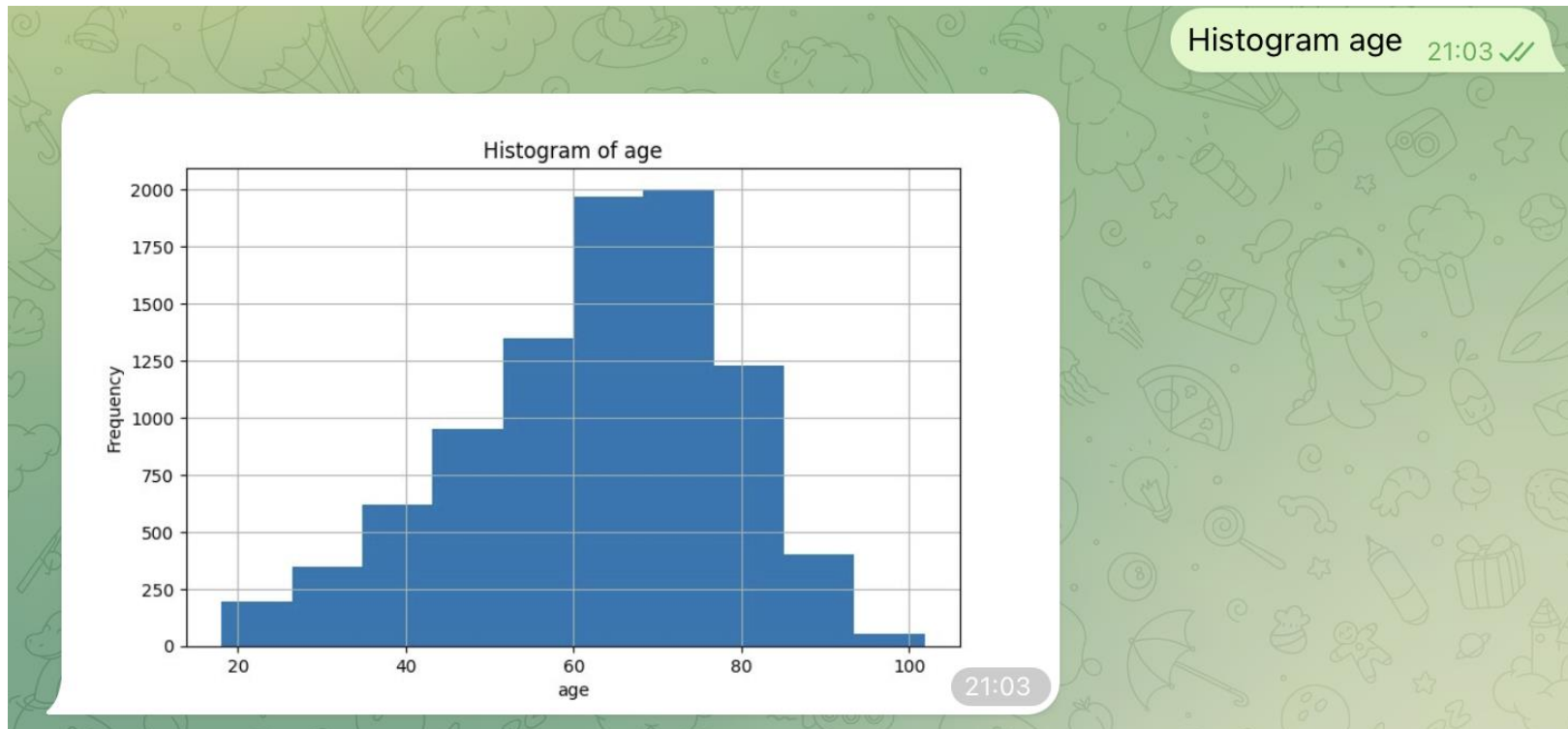
Dataset Histogram

Functionality:

This function creates a histogram for a selected numeric column.

Example:

The user presses Histogram, then types a column name, for example:



Search Column

Functionality:

This function allows the user to look for a column by available columns list

Example:

If the dataset has columns:

Available columns:

age
death
sex
hospdead
slos
d.time
dzgroup
dzclass
[num.co](#)
edu
income
scoma
charges
totcst
totmcst
avtisst
race
sps
aps
surv2m
surv6m

Search Column 21:06 ✓✓

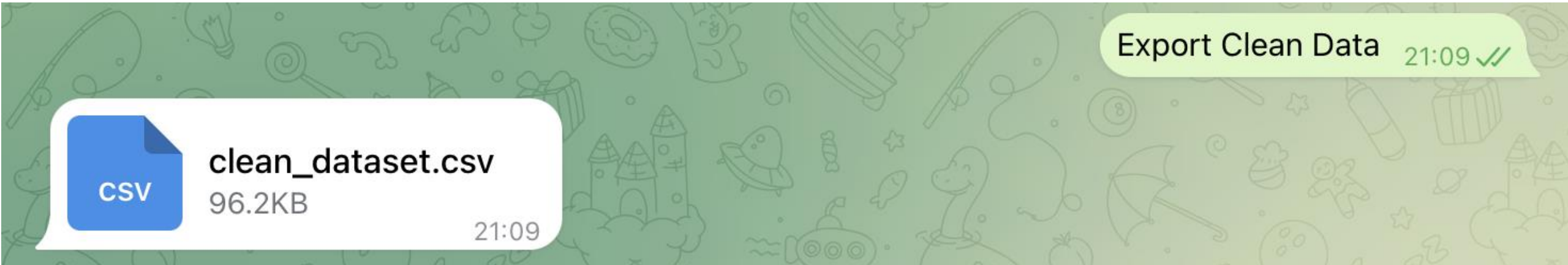
Export Clean Data

Functionality:

This function creates a cleaned version of the dataset and sends it back to the user as a file.

What the bot can do:

The bot can replace missing values or prepare a cleaner version of the dataset for download.



Help Function

Functionality:

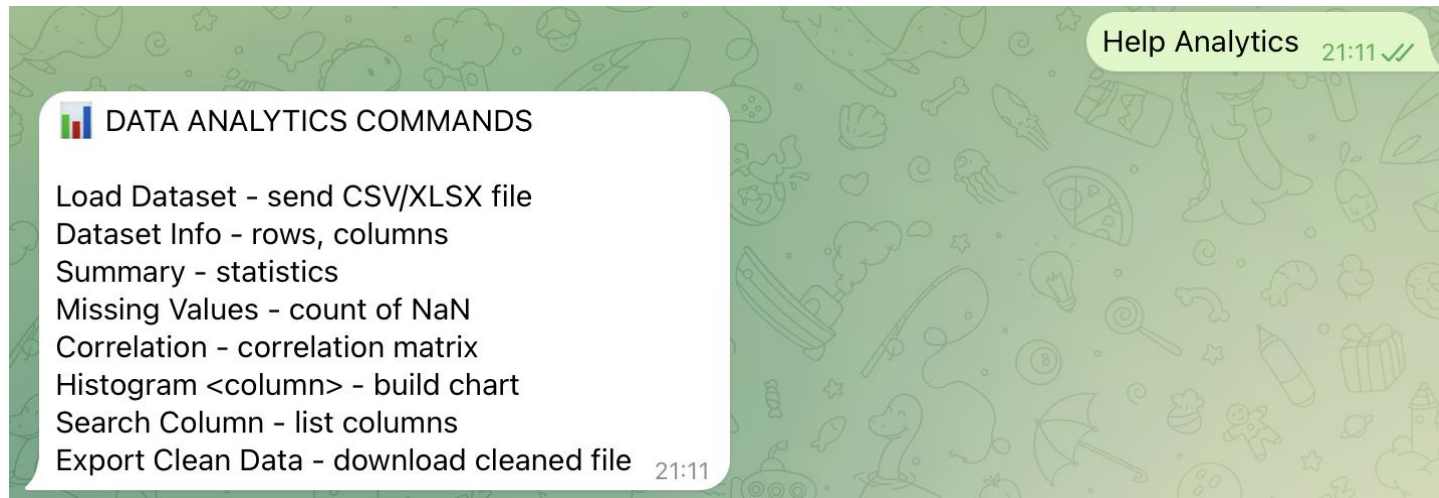
This function explains how to use the Data Analytics module.

The help message includes:

- available buttons;
- short description of each function;
- examples of user input;
- reminder to upload a dataset first.

Why it is useful:

It makes the bot more user-friendly and helps new users understand the available functions.



Fun Work Part

This module was added to make the bot more interactive. It shows that Telegram bots can work not only with files and datasets, but also with user actions, buttons, stickers, timer functions, and location messages.

Programming concepts used:

- Telegram message handlers
- Button menus
- User states
- Dictionaries
- Conditional statements
- Simple calculations
- Timer logic
- Location message handling
- Sticker sending
- Error handling with try/except

The part consists of 4 internal functions such as:



Calculator



Timer



Location



Sticker

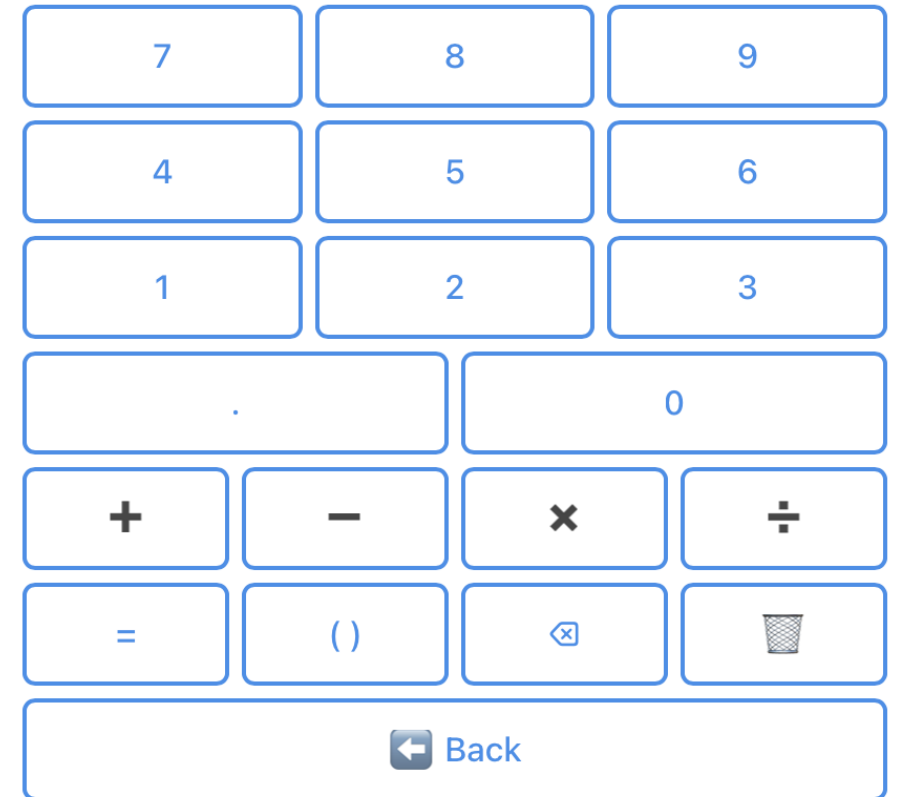
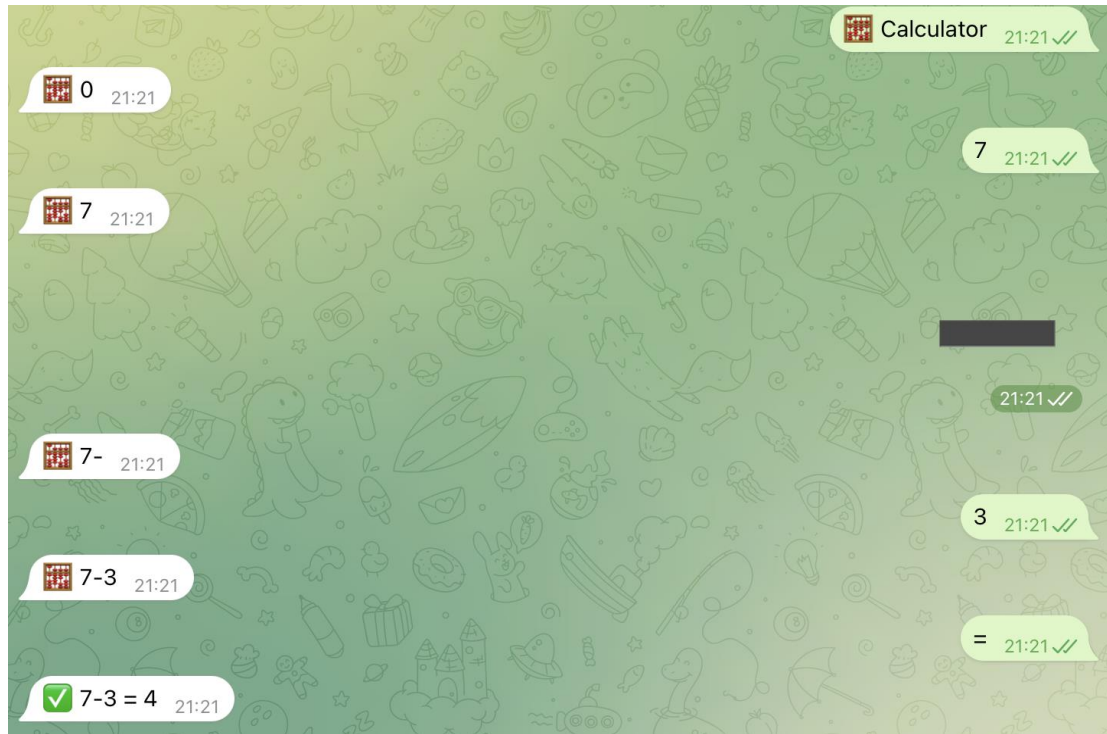


Back

Calculator

Functionality:

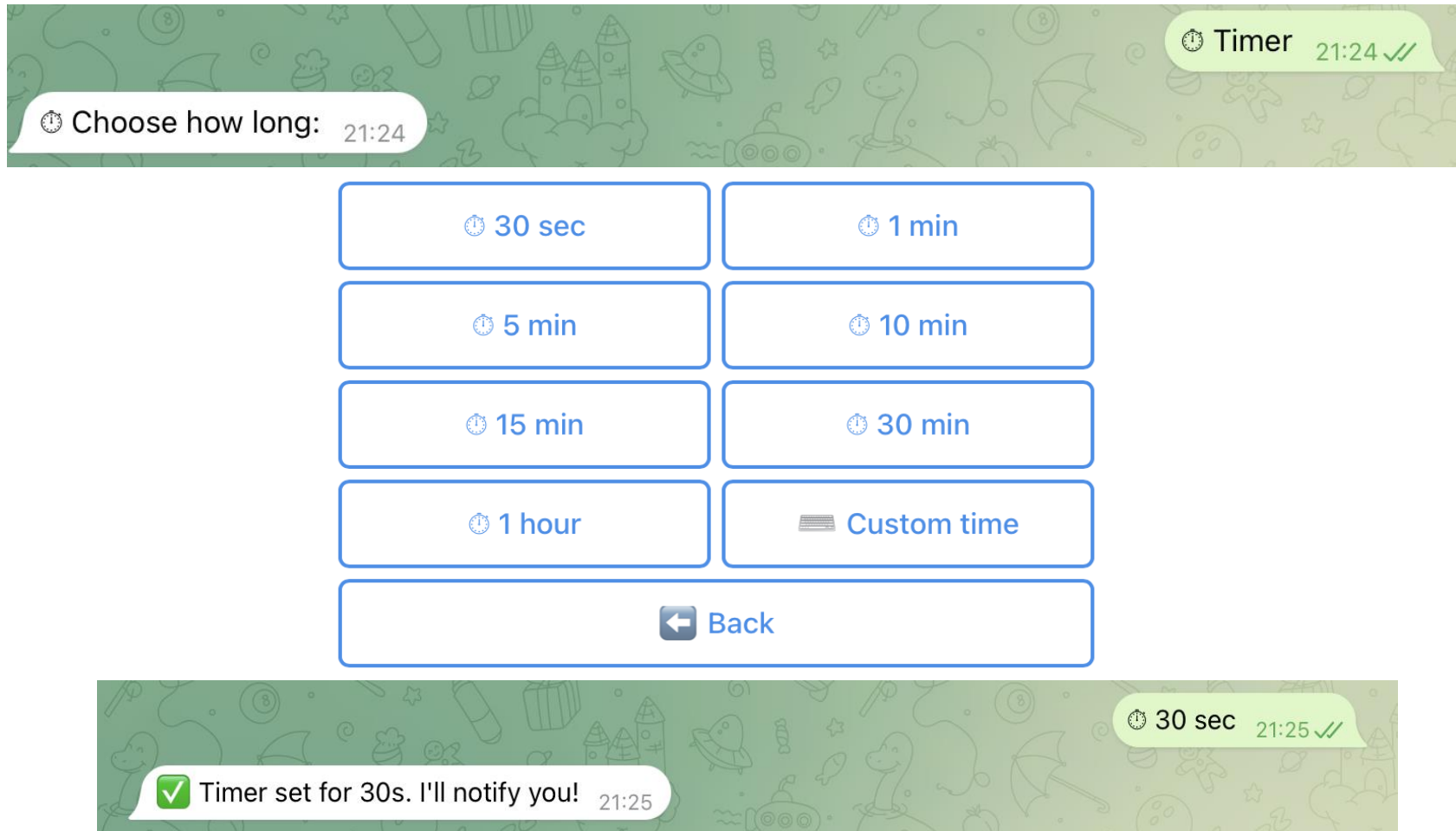
The calculator allows the user to enter a simple mathematical expression and receive the result directly in Telegram.



Timer

Functionality:

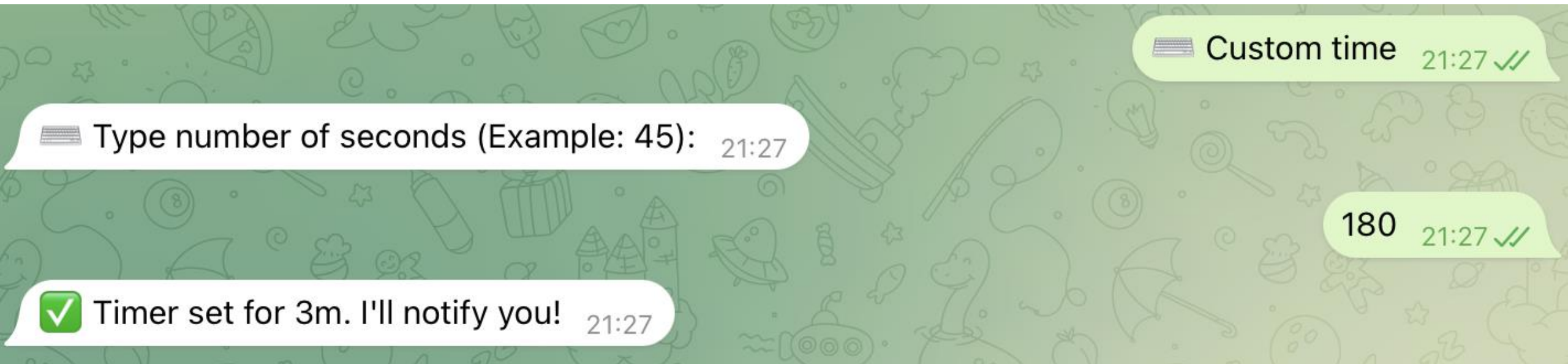
The timer allows the user to set a simple reminder for a selected time.



Custom Time

Functionality:

The custom timer allows the user to enter their own number of seconds or minutes.

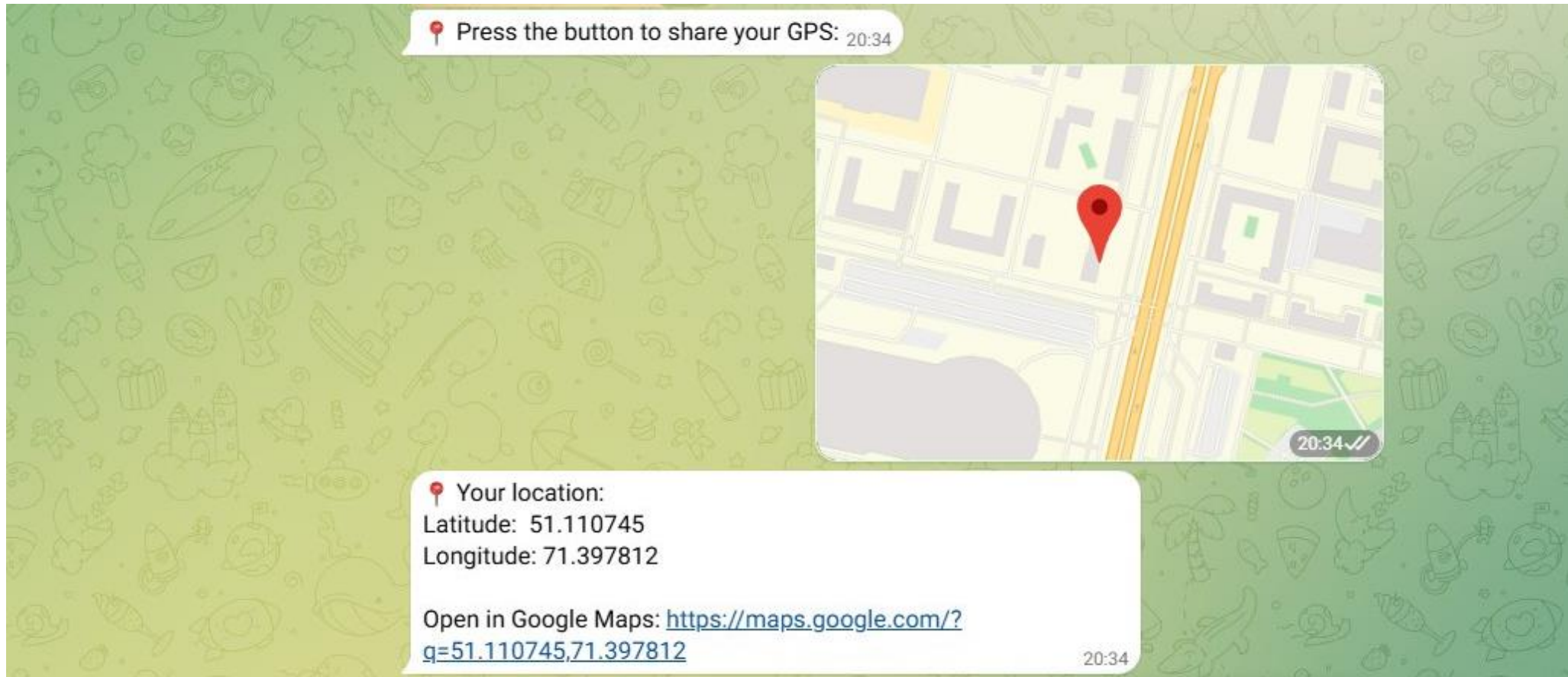


Location Function

Functionality:

The location function allows the user to share GPS coordinates with the bot.

After receiving coordinates, the bot can create a Google Maps link using latitude and longitude.



Sticker Function

Functionality:

The bot can send stickers when the user opens specific functions or performs certain actions.

Example:

When the user sets his timer, sticker will be sent immediately.





Thanks for Attention

May 11, 2026